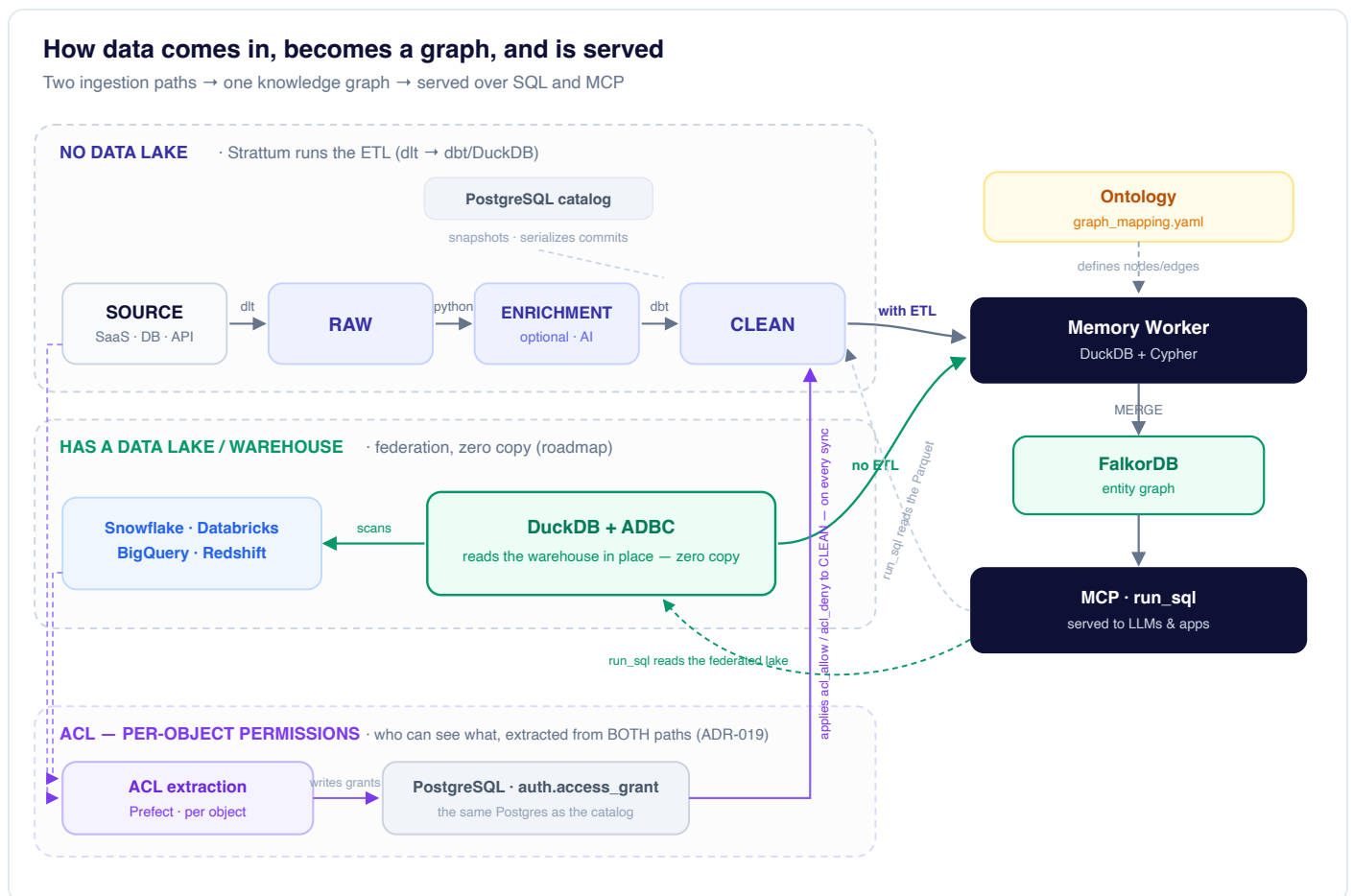


Architecture — Strattum Lakehouse (open lake)

🇧🇷 Versão em português: [ARQUITETURA-LAKEHOUSE.md](#)

Status: current (reference for the production architecture) · **Scope:** how data comes in (connectors), becomes tables (lake), gets modeled (dbt clean) and becomes a graph (memory-worker).

Three-line summary: each **connector** pulls data from a source and writes to the lake's **raw**; **dbt** transforms **raw** → **clean**; the **memory-worker** reads **clean** and builds the **graph**. The lake is **DuckLake** (Postgres catalog + Parquet on MinIO/S3), accessed through a single point: **LakeStore**.



Mental model (read this first)

The lake has **five boxes** and **one destination**:

SOURCE → RAW → (ENRICHMENT) → CLEAN → GRAPH — and everything is **served over SQL and over MCP**.

- **RAW** — the data as it came from the source, in **Parquet** in the lake (one table per resource).
- **CLEAN** — the **modeled** data (typed, deduplicated, with `external_id`). It is what the UI and the graph consume.
- **GRAPH** — entities and relationships in **FalkorDB**, built from CLEAN through an **ontology**.

Two entry paths — the only fork that matters:

Path	When	What Strattum does
With ETL	customer without a data lake	runs the pipeline: connector pulls → <code>RAW</code> → dbt → <code>CLEAN</code>
No ETL (federation)	customer with a lake/warehouse (Snowflake, BigQuery, ...)	no copying — DuckDB + ADBC reads the data <i>in place</i> and feeds the graph directly (roadmap)

The two paths converge on the **same graph** and are served by the **same** `run_sql` /MCP. The rest of this document details the **with-ETL** path (what runs in production today); federation is in §7 and in ADR-020.

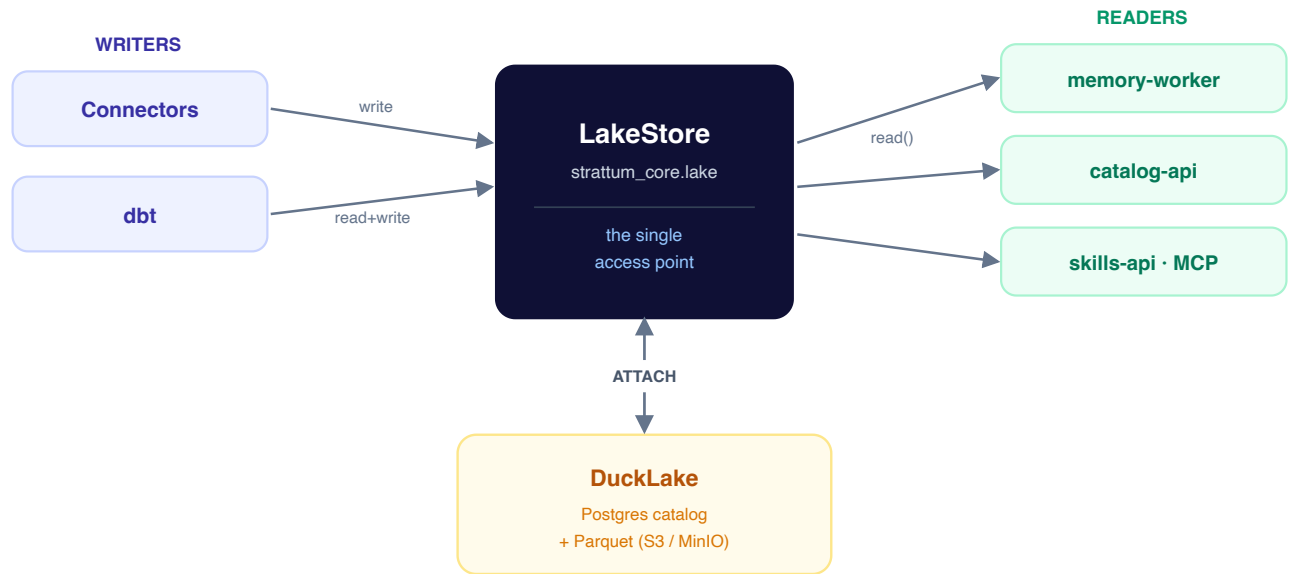
Where to touch the code: new connector → `services/pipelines/src/connectors/<c>/` + `flows/<c>_sync.py` · modeling → `dbt/models/clean/` · graph ontology → `graph_mapping.yaml` . Lake access is **always** via `strattum_core.lake.LakeStore` — never hand-written lake SQL.

The LakeStore — the heart of the lake

Before, each service opened the lake its own way (shared `.duckdb`, `write_delta`, Parquet globs). Today **everything goes through a single class** — `strattum_core.lake.LakeStore` . Connectors **write**, dbt reads and writes, memory-worker / catalog-api / skills-api / MCP **read** — all through here. It is the only place that knows the lake is DuckLake.

LakeStore — one access point for the whole lake

Connectors and dbt write; memory-worker, catalog-api and MCP read — all through the same class.



How it opens the lake (on every connection): an in-memory, stateless DuckDB (there is no `.duckdb` file anymore) that **(1)** loads the `ducklake + httpfs + postgres` extensions, **(2)** creates the S3 secret (MinIO/AWS/R2 — only the `ENDPOINT` changes) and **(3)** runs `ATTACH 'ducklake:postgres:<dsn>' AS lake (DATA_PATH 's3://...')`. Done: `lake.raw.<t>`, `lake.clean.<t>` resolve. Because the **catalog is Postgres**, many processes write at the same time (snapshot isolation) — that is what killed the single-writer lock.

The whole API (small on purpose):

Write	Read	Utility
<code>write(layer, table, data, mode, pk)</code>	<code>read("<layer>/<table>", watermark=...) → dict s</code>	<code>list_tables</code> · <code>table_exists</code> · <code>get_columns</code>
<code>write_records_batched(...)</code> — streaming in 10k batches (never holds the whole resource in RAM)	<code>connection()</code> — escape hatch for custom SQL (catalog-api, skills-api)	<code>delete_where_in (chunked)</code> · <code>bootstrap</code>

- `mode` = `overwrite` (default) · `append` · `merge` (upsert by `primary_key`). DuckLake has no PK/UNIQUE → merge is *delete-then-insert*; schema drift is tolerated (`INSERT ... BY NAME`).
- `read()` is **origin-agnostic** — it returns `dict s` whether the source is our `clean` or (future) a federated warehouse via ADBC. The memory-worker receives the same thing → **the graph path never changes**.
- Swapping formats** (DuckLake → Delta) = reimplementing this class; callers do not change.
- `attach_federation()` is the federation hook (task 03): since `read()` is already agnostic, the customer's lake/warehouse enters through the **same** reader — it is the bridge to the "no ETL" path in the diagram above.

1. The lake (DuckLake)

An **open** lakehouse: the **catalog** (schema, snapshots, file map) lives in **Postgres**; the **data** is **Parquet** on **S3/MinIO**. DuckDB is just the engine (embedded). Three layers, addressed as `lake.<layer>.<table>`:

- **raw** — the data as it came from the source, one table per resource (`raw."<connector>__<resource>"`).
- **enrichment** — AI columns (optional, additive).
- **clean** — the modeled layer (typed, `external_id`) that the graph and the UI consume.

2. How a connector brings data in

- **StratumConnector interface:** `authenticate · discover · extract(resource, state, *, limit) · test_connection . extract` **streams** (paginates / `fetchmany` / `yield` in a loop) — **never** `.fetchall()`; the resource never fits in RAM.
- **Incremental — one engine:** `sync_resource_to_raw` (`connectors/utils/resource_config.py`) reads the resource config from `connector_state` and decides: **incremental** → `cursor > last_value` , **MERGE by PK**, advances the watermark · **full_refresh** (safe default) → pulls everything and overwrites.
- **PK and cursor:** SaaS = a fixed fact of the API; databases = the user picks in the UI (stored in `connector_state`). The flow reads them with `get_resource_configs("<c>")` .
- **Consolidated tables** (`notion__pages` , `slack__messages` , ...) — N sub-resources in a single table → **always merge** , never `overwrite` (or you erase the siblings).

3. The flow (single pattern)

Every `flows/<c>_sync.py` has the same shape (ref: `asaas_sync.py` , `bigquery_sync.py`) — only the extraction core changes:

```
load_<c>_credentials → load_<c>_config → _build_connector(creds)
    → discover (ONE test_connection) → extract_<c>_resource (per resource, honors the mode)
    → run_<c>_dbt → thin @flow
```

The `@flow` name = the file name (`deploy.py` discovers by glob). Every flow carries `@with_sync_progress` (writes `running/completed/failed` to `connector_sync_progress`).

4. `raw → clean` (dbt) and enrichment

- **dbt clean** reads `lake.raw."<c>__..."` . `run_dbt_for_connector("<c>")` **auto-discovers** the models that depend on the connector and runs only those (skips 0-row ones). In-process (`dbtRunner`).
- **Concurrency (advisory lock):** a *cross-source* model (e.g. `clean.customers` fed by postgres **and** mongodb) is discovered by both flows and would collide on the DuckLake commit. `_run_dbt` **serializes per model** with a Postgres advisory lock (`connectors/utils/locks.py`): 1 run per model at a time, fail-open, kill-switch `DBT_LOCK_DISABLED` . Detail in [exp 09](#).
- **ACL:** every clean model carries `acl_allow / acl_deny` (macro `strattum_acl_columns()` — ADR-019). Permissions are **extracted per object from both paths** (with-ETL **and** federated sources) by a Prefect task and written to `auth.access_grant` — **in the same Postgres as the catalog**; a recurring process **applies** the columns onto clean on every sync.

- **Databases:** clean generated at runtime (catalog-api synthesizes the SELECT: PK→ `external_id` , PII `drop / hash`).
- **Enrichment (optional):** with AI transforms it runs `raw → enrichment → clean` ; otherwise `raw → clean` directly.

5. `clean → graph`

Only for connectors with an `ontology_fragment.yaml` (the fixed-schema SaaS ones). The fragment declares nodes/edges/timeline/ER; it is merged into `graph_mapping.yaml` ; the **memory-worker** reads `lake.clean` (via `LakeStore.read`), mints `entity_id` (deterministic entity resolution — the same key, e.g. an email, becomes the same id across runs, unifying sources), generates Cypher (MERGE) and writes to **FalkorDB** (graph `strattum_memory`). Dynamic-schema connectors (airtable + databases) have **no** fragment — the FDE maps per customer post-onboarding.

How `graph_mapping.yaml` gets created — two ways, same versioned result:

- **Fixed-schema connector** → the `ontology_fragment.yaml` ships with the connector; the FDE reviews it and merges it into the unified `graph_mapping.yaml` .
- **Dynamic-schema (databases, warehouse, airtable)** → there is no factory fragment; the map is **written per customer**:
- **Via the API** (FDE / automation): `PUT /v1/ontology` (saves a version) + `POST /v1/ontology/apply` (applies; optional `reset=true` to clear ghost entities from old ontologies — ADR-008).
- **Via the UI** (the customer): **Settings** → **Ontology**. The **Graph** tab shows the map as a diagram; the **YAML** tab is the editor (Monaco) — **Edit** → (validates the YAML and the columns against the real `clean` tables) → **Save and apply**, versioned with history and rollback.

Ontologia
Configure os tipos de entidade, relacionamentos e mapeamento de fontes.

Gráfico **YAML**

Versão 4 **Atual** Editar

```

1  version: "1"
2
3  # Benchmark ontology: Users, Contracts and Products from the DuckLake clean layer.
4  # Graph node sources are BOUNDED slices (clean/graph_users, clean/graph_contracts)
5  # so the graph fits the 3.8 GB Docker VM; clean.users / clean.contracts hold the
6  # full 2M / 1M for the catalog + throughput numbers. Contract parties resolve to
7  # Users by the postgres user id (processante_id / processado_id -> user_id).
8
9  nodes:
10   - label: User
11     source: clean/graph_users
12     connector: postgres
13     id_field: user_id
14     er_fields: [email]
15     display_tier: entity
16     display_color: blue
17     properties: [user_id, name, email, cpf, city, updated_at]
18
19   - label: Product
20     source: clean/products
21     connector: mongodb
22     id_field: product_id
23     er_fields: []
24     display_tier: entity
25     display_color: green
26     properties: [product_id, product_name, product_category]
27
28   - label: Contract
29     source: clean/graph_contracts
30     connector: mongodb

```

Ver histórico de versões

In the [benchmark](#) the `User / Contract / Product` ontology was created exactly through this path (via the API, registered as **Version 4**) — the screenshot above is that ontology, open in a customer's UI.

6. Connector summary

Incremental legend: ✅ working (flow passes the state, merge by PK, watermark advances) · 💎
CDC/webhook · ⚠️ caveat (see note). **Graph legend:** ✅ ships an `ontology_fragment.yaml` (graph map ready
out of the box) · 📝 the **FDE** writes the ontology **per customer** (the schema belongs to the customer — a
fixed map cannot be shipped).

Connector	Type	How it extracts (streaming)	Incremental	Graph
asaas	fixed SaaS	REST offset/limit + <code>hasMore</code>	✓ cursor <code>dateCreated</code> / <code>transferDate</code>	✓
bigquery	warehouse	SQL <code>fetchmany(1000)</code> ; <i>federation is the future</i>	✓ cursor via <code>connector_state</code>	🔥 FDE
postgres	database	SQL <code>stream_results</code> + <code>fetchmany</code>	✓ cursor via <code>connector_state</code>	🔥 FDE
mysql	database	SQL <code>stream_results</code> + <code>fetchmany</code>	✓ cursor via <code>connector_state</code>	🔥 FDE
mongodb	database	cursor <code>.batch_size</code>	✓ <code>cursor_field</code> via <code>connector_state</code>	🔥 FDE
airtable	dynamic	REST + webhook CDC	🔹 webhook (<code>changed_record_ids</code>) + delete	🔥 FDE
salesforce	fixed SaaS	SOQL <code>query_all_iter</code> (streaming)	✓ <code>LastModifiedDate</code> , PK <code>Id</code>	✓
hubspot	fixed SaaS	REST paging (associations in batches)	✓ <code>hs_lastmodifieddate</code>	✓
jira	fixed SaaS	REST <code>nextPageToken</code>	✓ <code>updated</code> (+ walk-cursor for links/comments)	✓
zendesk	fixed SaaS	REST incremental exports (cursor)	✓ <code>updated_at</code> (→ <code>start_time</code>)	✓
slack	fixed SaaS	REST cursor paging	✓ <code>ts</code> , PK <code>channel:ts</code>	✓
clickup	fixed SaaS	REST page paging	✓ <code>date_updated</code> (tasks)	✓
confluence	fixed SaaS	REST start/limit	✓ <code>version_when</code> (merge per space)	✓
notion	dynamic	REST <code>start_cursor</code>	✓ <code>last_edited_time</code> (merge per database)	✓
google_analytics	fixed SaaS	<code>run_report</code> paginated	✓ <code>date</code> (MERGE-by-day)	✓ (<code>nodes:</code> <code>[]</code>)
microsoft365	SaaS/files	per-drive streaming (<code>@odata.nextLink</code>)	✓ per-drive cursor, merge <code>doc_id</code>	✓
google_drive	SaaS/files	streaming listing + content	⚠ <code>modified_at</code> wired, but the watermark does not persist (pre- existing bug — §8)	✓

⚠️ The 🧑‍🔧 FDE in the Graph column does not mean "does not reach the graph"

Every connector **can** feed the graph. The difference is **who writes the map** (the ontology linking `clean` → nodes/edges):

- **Fixed schema (SaaS)** — the connector already **knows** its entities (a HubSpot "deal" is always a deal), so it **ships** a factory `ontology_fragment.yaml` → graph **out of the box** (✅).
- **Dynamic schema (databases `postgres` / `mysql` / `mongodb`, warehouse, `airtable`)** — the schema is **defined by the customer**; Strattum cannot know in advance which tables/columns exist, so shipping a fixed map is **impossible**. There, the **FDE writes the ontology per customer**, after onboarding, looking at real data (🧑‍🔧).

Living proof — this is exactly the benchmark's case. The **LAKEHOUSE BENCHMARK** runs `postgres` + `mongodb` (both 🧑‍🔧) **all the way to the graph** — 290k nodes, 300k edges — precisely because the *bespoke* ontology (`graph_mapping.yaml`: `User` / `Contract` / `Product` nodes + `PROCESSANTE` / `PROCESSADO` / `SOBRE` edges) was hand-written for that customer. The 🧑‍🔧 is **exactly that step** — not a limitation.

How each one is organized — every connector has the same layout: `auth.py` · `config.py` · `connector.py` · `schemas/` · `transforms/` (+ `transforms.yaml`) · `acl.py` · `ontology_fragment.yaml` + `knowledge_fragment.yaml` (fixed-schema only) · `tests/`. Databases omit transforms/fragments (the schema belongs to the customer, generated at runtime; the ontology comes from the FDE).

6.1 Merge keys (PK) and cursors

In DuckLake **there is no PRIMARY KEY** — the "merge" is *delete-then-insert* by the key below. The **PK** identifies the row (the upsert deletes the key and reinserts in place, keeping `raw` as a deduplicated snapshot); the **cursor** is the date column the incremental compares (`cursor > last_value`) and persists as the watermark. Where the two come from:

- **Fixed-schema SaaS** — PK and cursor are a **fact of the API**, constants in the connector's `config.py` / `schemas/`; the user does **not** choose.
- **Databases and warehouse** (`postgres`, `mysql`, `bigquery`, `mongodb`) — PK and cursor are **chosen by the user per table** in the UI (column picker) and stored in `connector_state.__config.resources`. MongoDB assumes `_id` as the default.
- **Synthetic** — `slack` composes `message_key = channel_id:ts` (`ts` alone is not unique across channels); `airtable` uses the API's own `record_id`.

Connector	PK (merge key)	Cursor (incremental)	PK origin
asaas	id	dateCreated · transferDate (transfers)	API const
salesforce	Id	LastModifiedDate	API const
hubspot	id	hs_lastmodifieddate	API const
jira	id (users: account_id)	updated (+ walk issue_updated)	per-resource dict
zendesk	id	updated_at	API const
clickup	id	date_updated (tasks)	API const
confluence	id	version_when	API const
notion	id	last_edited_time	API const
slack	message_key (=channel_id:ts)	ts	synthetic
google_analytics	date	date	API const
microsoft365	doc_id	modified_at	const
google_drive	doc_id ¹	modified_at ¹	const
airtable	record_id (comments: comment_id)	CDC webhook (changed_record_ids)	API const
postgres	chosen in the UI	chosen in the UI	connector_state
mysql	chosen in the UI	chosen in the UI	connector_state
bigquery	chosen in the UI	chosen in the UI	connector_state
mongodb	_id (default)	chosen in the UI (cursor_field)	connector_state

¹ google_drive: PK/cursor wired, but the watermark **does not persist** today (§8) → effectively full on every run.

Resources that are always full (never incremental — supports_incremental=False, the flow forces full_refresh regardless of what the UI asks): asaas subscriptions; clickup members / lists / spaces; hubspot *_associations (composite pair (from_id,to_id) with no single-column PK, and a merge would not detect a removed association); jira projects / users and comments; zendesk comments.

Consolidated tables — always merge, never overwrite (N sub-resources written into a single table; overwrite would erase the siblings): notion__pages (merge per database), confluence__pages (merge per space), slack__messages (merge per channel).

6.2 What each connector needs (credentials)

Fields read in each connector's authenticate() — what the customer fills in the config modal for the connection to work:

Connector	Credentials / minimum config
asaas	api_key + environment (sandbox/production)
salesforce	username , password , security_token , client_id , client_secret
hubspot	access_token (Private App)
jira	domain , email , api_token (+ optional custom_field_ids)
zendesk	subdomain , email , api_token
clickup	api_key + team_id
confluence	domain , email , api_token (+ space_keys)
notion	token (integration) + database_ids
slack	bot_token + channel_ids
google_analytics	service_account_json + property_id
microsoft365	tenant_id , client_id , client_secret (+ selected_drives / selected_sites)
google_drive	service_account_json + folder_ids / file_ids (+ subject_email for domain-wide)
airtable	AIRTABLE_PAT + AIRTABLE_CONFIG (selected bases/tables)
postgres / mysql	connection_string
bigquery	project_id + service_account_info
mongodb	uri

7. The 3 reference archetypes

- **airtable** — **dynamic-schema / CDC**: user-selectable strategy, webhook, delete_where_in , opaque raw (fields_json), no ontology.
- **asaas** — **fixed-schema SaaS / cursor**: incremental via connector_state , cursor from the ResourceSchema , has an ontology.
- **bigquery** — **warehouse**: streaming via fetchmany , cursor via connector_state ; the future target is **federation** (see ../strattum-brain/docs/adr/020-bigquery-federation-vs-etl.md).

8. Known follow-ups (post-reform)

All 17 connectors were brought to the standard (real incremental, streaming, uniform flow shape, connector+flow+E2E tests). What remains:

1. **Per-resource sync-mode UI** — done for **5** (airtable, asaas, clickup, hubspot, zendesk). **6** SaaS remain (salesforce, google_analytics, jira, slack, confluence, notion): resource-config route in catalog-api + switches in the modal + i18n (pattern already reproduced, just replicate). microsoft365 / google_drive have no toggle (1 resource / selection via YAML).
2. **google_drive watermark** — extract_google_drive_source calls conn_logger.finish(new_state=None) , so the modified_at cursor never persists → re-scan on every run. Needs to track the max and persist via new_state .

- 3. **catalog-api** `_install_shipped_transforms` — the `_DOC_CONNECTOR_TYPES` gate needs to include the SaaS connectors that now ship clean SQL (e.g. `google_analytics` gained transforms).
- 4. **E2E test hygiene** — some E2E pytest tests monkeypatch `get_lake / _attach_sql` at module level without restoring → 1 flake under combined execution (passes in isolation). Restore via fixture.
- 5. **FDE reviews (ontology fragment)** — agent flags: `salesforce` (`HAS_DEAL / OPENED_CASE` join_key on the wrong node), `slack` (`Agent` node without a role filter, edges without `source:`), `clickup` (`TaskList / Space` nodes without a transform).
- 6. **Pre-existing bugs found (not introduced)** — `mysql` (mismatch `MYSQL_CLIENT_URL ≠ connection_string` in auth), `confluence` (the modal's `spaces` dropped by `SaveConnectorRequest` → selection never persists), `hubspot` (`test_connection` raises instead of returning `False`). The dead `QUALIFY ... _dlt_load_id` in the clean SQL (`jira/notion/confluence`) **was already fixed** in the reform.

Standardized per-connector reform: `/reform-connector <name> (.claude/commands/)`.

9. Live E2E validation (DuckLake migration) — 2026-07-19

Closing the migration to the **open lake**: run the real stack end to end and prove that data from **two distinct sources** lands in a **single clean table**, with real incremental and Parquet on MinIO. Not a simulated test — it is the local production stack (Postgres catalog + MinIO + Prefect worker + console UI).

Setup: two connectors created **through the UI** (`postgres`, `mongodb`), each with 4 `customers` at the source. Scheduled sync (hourly cron) + manual triggers.

Proven flow (end to end): UI → connector → sync → `raw` (Parquet on MinIO, both sources) → dbt auto-discovery → unified `lake.clean.customers` → **FalkorDB graph** (memory-worker) → read via **MCP** `run_sql`.

Before → after:

Signal	Before (pre-migration)	After (measured)
Storage	shared <code>.duckdb</code> / local delta files	DuckLake : Postgres catalog + Parquet on MinIO (<code>s3://strattum-lake</code>)
<code>raw</code>	per-connector, engine-coupled	<code>raw.postgres__public__customers = 4,</code> <code>raw.mongodb__demo__customers = 4</code>
Unified <code>clean</code>	nonexistent (1 source per table)	<code>clean.customers = 8</code> (<code>postgres=4 + mongodb=4</code>), email <code>LOWER/TRIM</code> , per-source <code>external_id</code>
Incremental	full on every run	MERGE by PK : Ana's email update (<code>ana@</code> → <code>ana.nova@</code>) landed without duplicating (1 row); mongo picked Gustavo up as the delta (3→4)
Watermark	—	advances and persists in <code>connector_state</code> (<code>postgres 2026-07-05</code> , <code>mongo 2026-07-04</code>)
Concurrent writes	single-writer (file lock)	2 connectors write to the same lake at the same time (Postgres catalog)

clean → graph (memory-worker): a minimal ontology authored for `clean.customers` (dynamic-schema ships no fragment — the FDE maps per customer): `Customer` node, `source: clean/customers`, `id_field:`

`email` + `er_fields: [email]` (the ER key **is** the email, with no source namespace → two rows with the same email collapse into a single node across sources). Ran the pipeline directly → **8 customer nodes** in FalkorDB (`strattum_memory`), deterministic `entity_id = uuid5(email)`, `errors: []`. (The demo's 8 emails are distinct, so there is no collapse to demonstrate in this dataset — the ER is wired; it just needs a shared email to display the unification.)

`graph → read (MCP run_sql)`: proved the path `MCP run_sql → skills-api /v1/skills/sql → LakeStore` → **DuckLake**: a `SELECT OVER lake.clean.customers` returns the 8 unified customers (markdown, 500-row cap); `DELETE / UPDATE / DROP / read_parquet(glob)` **blocked** (400). `skills-api` came up host-side with the same auth bypass as `catalog-api`.

Bugs found in the E2E → fixed: 1. **Concurrent dbt (same model)**: two syncs on the same tick auto-discover `clean.customers` and collide on the DuckLake commit — reproduced live (the 01:00 run failed postgres's dbt with `Failed to commit DuckLake transaction`, recovering only on retry). **Fix**: per-model advisory lock (§4) + concurrency ceiling of 1 on the Prefect work pool (`STRATTUM_DATA_WORK_POOL_CONCURRENCY`, converged in `flows.deploy`). **Live re-test**: 2 syncs at the same instant → postgres locked and materialized, mongo **blocked for 1.6s** and ran after the release — **0 collisions, 0 retries**. Coverage:

`connectors/utils/tests/test_locks.py` (11 tests). 2. **memory-worker × typed timestamp**: a `TIMESTAMPTZ` column from `clean` comes back from DuckDB as `datetime`, and FalkorDB's parameter serializer inlines it as a raw literal (`Invalid input '-'`) → **all** nodes in a batch fail (affects any `clean` with a date column). **Fix**: parameter sanitization in `memory_worker/falkordb_client.py` (`datetime/date/Decimal/UUID/bytes → primitive`) at the single `execute / execute_batch` entry point. Coverage: `tests/test_falkordb_params.py` (9 tests).

E2E follow-ups (outside the migration's own scope): demonstrate the ER collapse with an email shared across sources; the *incremental clean layer* toggle in the UI is still a no-op; mongo's string cursor has a type edge (BSON `$gt`) to harden — the last two already tracked as their own tasks.